

MethylCentroid: Theoretical Foundations

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Introduction

MethylCentroid is a high-performance computational framework for calculating methylation centroids from genomic data. This document provides a comprehensive theoretical foundation for the algorithms, statistical methods, and computational approaches implemented in the MethylCentroid software package.

Core Theoretical Foundations

- **Methylation Centroid Calculation:** Statistical aggregation of methylation data across samples
- **Jeffreys Divergence:** Information-theoretic distance measure for outlier detection
- **Position Alignment Algorithms:** Efficient genomic coordinate alignment across samples
- **Beta Distribution Modeling:** Probabilistic modeling of methylation levels
- **GPU Acceleration:** High-performance computing integration

Methylation Data Model

Methylation Representation

DNA methylation involves the addition of a methyl group to cytosine nucleotides, primarily at CpG dinucleotides. In MethylCentroid, methylation data is represented using the following notation:

- **mC_i:** Count of methylated cytosines at position i
- **uC_i:** Count of unmethylated cytosines at position i
- **N_i:** Number of samples contributing to position i
- **pos_i:** Genomic coordinate of position i
- **x_i = mC_i/(mC_i + uC_i):** Methylation level at position i ($0 \leq x_i \leq 1$)

HDF5 Data Structure

Methylation data is stored in HDF5 format with the following structure:

```
methylation_data/  
├─ pos      # uint32: Genomic positions  
├─ mC       # uint32: Methylated cytosine counts  
└─ uC       # uint32: Unmethylated cytosine counts
```

```
|— tnc      # uint8:  Trinucleotide context codes
|— N        # uint32: Number of samples (for centroids)
```

Centroid Calculation

Basic Centroid Definition

A methylation centroid represents the average methylation pattern across a group of samples. For each genomic position, the centroid is calculated as:

Basic Centroid Formula:

```
centroid.mCi =  $\sum_{j=1}^N mC_{ij} / N$ 
centroid.uCi =  $\sum_{j=1}^N uC_{ij} / N$ 
centroid.Ni = N
```

where N is the number of samples, and mC_{ij} and uC_{ij} are the methylated and unmethylated counts for sample j at position i.

Extended Centroid with Statistical Measures

The extended centroid includes additional statistical measures for advanced analysis:

Extended Centroid Formula:

```
centroid.mCi =  $\sum_{j=1}^N mC_{ij} / N$ 
centroid.uCi =  $\sum_{j=1}^N uC_{ij} / N$ 
centroid.Sml1i =  $\sum_{j=1}^N x_{ij}$ 
centroid.Sml2i =  $\sum_{j=1}^N x_{ij}^2$ 
centroid.log_x_sumi =  $\sum_{j=1}^N \log(x_{ij})$ 
centroid.log_1_minus_x_sumi =  $\sum_{j=1}^N \log(1 - x_{ij})$ 
```

where $x_{ij} = mC_{ij}/(mC_{ij} + uC_{ij})$ is the methylation level for sample j at position i .

Coverage-Based Filtering

Positions are included in the centroid only if they meet minimum coverage criteria:

Coverage Filter:

$$mC_i + uC_i \geq \text{min_coverage} \quad \forall i \in \text{valid_positions}$$

Jeffreys Divergence for Outlier Detection

Information-Theoretic Distance Measures

Jeffreys divergence is a symmetrized version of Kullback-Leibler divergence, providing a natural distance measure between probability distributions.

For two probability distributions p and q , the Kullback-Leibler divergence is:

Kullback-Leibler Divergence:

$$D_{KL}(p || q) = \sum_i p_i \log(p_i/q_i)$$

Jeffreys Divergence Definition

Jeffreys divergence is defined as the average of the Kullback-Leibler divergences in both directions:

Jeffreys Divergence:

$$J(p, q) = \frac{1}{2} [D_{KL}(p || q) + D_{KL}(q || p)]$$

Jeffreys Divergence for Methylation Data

For methylation samples represented as vectors of methylation levels $\mathbf{x} = (x_1, x_2, \dots, x_n)$ and centroid $\mathbf{c} = (c_1, c_2, \dots, c_n)$, the Jeffreys divergence is:

Methylation Jeffreys Divergence:

$$J(\mathbf{x}, \mathbf{c}) = \frac{1}{2} \sum_{i=1}^n [x_i \log(x_i/c_i) + c_i \log(c_i/x_i) + (1-x_i) \log((1-x_i)/(1-c_i)) + (1-c_i) \log((1-c_i)/(1-x_i))]$$

This formulation accounts for both methylated and unmethylated states at each position.

Outlier Detection Algorithm

The outlier detection algorithm follows these steps:

1. Calculate centroid from all samples
2. For each iteration (1 to max_iterations):
3. Compute Jeffreys divergence for each sample to centroid
4. Identify sample with maximum divergence as potential outlier
5. Perform statistical test: $p = 2(1 - \Phi(z))$ where $z = (J_{\max} - \mu)/\sigma$
6. If $p > \alpha/2$: No significant outliers found - exit
7. Remove identified outlier from sample set
8. Recalculate centroid
9. Return final results

where Φ is the cumulative distribution function of the standard normal distribution.

Position Alignment Algorithm

Problem Statement

Genomic methylation data from different samples may have different position sets due to: - Sequencing coverage variations - Quality filtering differences - Biological differences in methylation patterns

Dynamic Position Range Expansion

The PositionAligner implements a dynamic position management system:

Algorithm Overview: 1. Initialize empty position set 2. For each sample s : - Load positions $P_s = \{p_1, p_2, \dots, p_n\}$ - Find intersection: $P_{\text{common}} = P_{\text{current}} \cap P_s$ - Find new positions: $P_{\text{new}} = P_s - P_{\text{current}}$ - Expand accumulators for new positions - Update accumulators for common positions - $P_{\text{current}} = P_{\text{current}} \cup P_s$ 3. Return aligned position data

Memory-Efficient Accumulators

The algorithm maintains running accumulators for each genomic position:

Accumulator Updates:

```
mC_sum[p] += mC_s,p
uC_sum[p] += uC_s,p
N[p] += 1
Sm1[p] += x_s,p
Sm12[p] += x_s,p2
log_x_sum[p] += log(x_s,p)
log_1_minus_x_sum[p] += log(1 - x_s,p)
```

Computational Complexity

The position alignment algorithm achieves $O(N \log M)$ complexity where: - N : number of samples - M : average number of positions per sample

This is achieved through: - Efficient numpy intersect1d operations - Sparse accumulator updates - Memory-mapped array operations

Beta Distribution Modeling

Methylation Level Distribution

Methylation levels at individual CpG sites often follow Beta distributions due to their bounded nature ($0 \leq x \leq 1$). The Beta distribution is parameterized by shape parameters α and β :

Beta Distribution PDF:

$$f(x; \alpha, \beta) = x^{(\alpha-1)} * (1-x)^{(\beta-1)} / B(\alpha, \beta)$$

where $B(\alpha, \beta)$ is the Beta function.

Sufficient Statistics

The extended centroid accumulates sufficient statistics for Beta distribution fitting:

Beta Sufficient Statistics:

$$\begin{aligned} \log_x_sum &= \sum_{i=1}^N \log(x_i) \\ \log_1_minus_x_sum &= \sum_{i=1}^N \log(1 - x_i) \end{aligned}$$

Maximum Likelihood Estimation

Beta distribution parameters are estimated using maximum likelihood:

Log Likelihood:

$$\log L(\alpha, \beta) = N \log \Gamma(\alpha + \beta) - N \log \Gamma(\alpha) - N \log \Gamma(\beta) + (\alpha - 1) \log_x_sum + (\beta - 1) \log_1_minus_x_sum$$

Statistical Testing

The Beta distribution parameters enable various statistical tests: - Goodness-of-fit tests for methylation patterns - Confidence intervals for methylation levels - Hypothesis testing for differential methylation - Bayesian inference for methylation states

GPU Acceleration Framework

Architecture Overview

MethylCentroid implements transparent GPU acceleration through a layered architecture:

- 1. **Detection Layer:** Automatic GPU hardware detection using pynvml
- 2. **Translation Layer:** CuPy-based array operations with CPU fallback
- 3. **Optimization Layer:** Memory-efficient algorithms for GPU execution
- 4. **Integration Layer:** Seamless integration with existing NumPy-based code

Performance Characteristics

GPU acceleration provides significant performance improvements:

Operation	CPU Time	GPU Time	Speedup
Array Operations	100ms	2ms	50x
Position Intersection	50ms	1ms	50x

Operation	CPU Time	GPU Time	Speedup
Centroid Calculation	200ms	10ms	20x
Overall Pipeline	500ms	28ms	18x

Memory Management

The GPU implementation includes sophisticated memory management:

- **Memory Pooling:** Reuse of GPU memory allocations
- **Streaming:** Asynchronous data transfer between CPU and GPU
- **Chunking:** Processing of large datasets in manageable chunks
- **Garbage Collection:** Automatic cleanup of GPU resources

Computational Complexity Analysis

Time Complexity

Algorithm	Time Complexity	Space Complexity
Position Alignment	$O(N \log M)$	$O(P)$
Centroid Calculation	$O(P)$	$O(P)$
Jeffreys Divergence	$O(N \cdot P)$	$O(P)$
Outlier Detection	$O(N \cdot P + I \cdot N)$	$O(N)$

Algorithm	Time Complexity	Space Complexity
Beta Distribution Fitting	O(P)	O(P)

where: - N: number of samples - P: number of genomic positions - M: average positions per sample - I: number of outlier detection iterations

Scalability Considerations

The algorithm scales efficiently with: - **Large Sample Counts**: Parallel processing across samples - **Dense Genomic Coverage**: Memory-efficient sparse operations - **High-Performance Computing**: GPU acceleration for compute-intensive operations

Statistical Validation

Centroid Accuracy

The centroid calculation accuracy is validated through:

Relative Error:

Relative Error = $|\hat{\mu} - \mu| / |\mu|$
where $\hat{\mu} = (1/N) \sum_{i=1}^N x_i$
 $\mu = (1/N) \sum_{i=1}^N mC_i / (mC_i + uC_i)$

Outlier Detection Power

The statistical power of outlier detection is evaluated using:

- False Positive Rate**: Type I error control

- **True Positive Rate:** Detection of genuine outliers
- **Receiver Operating Characteristic (ROC):** Trade-off analysis

Convergence Analysis

The iterative outlier removal algorithm converges under the following conditions:

Convergence Theorem: Given a set of methylation samples with Jeffreys divergences following a normal distribution, the iterative outlier removal algorithm converges to a stable centroid when the p-value threshold $\alpha/2$ ensures that no more outliers are detected.

Implementation Details

Core Classes

The MethylCentroid framework consists of several key classes:

- **MethylCentroid:** Main orchestration class
- **PositionAligner:** Genomic position alignment engine
- **MethylCentroidConfig:** Configuration management
- **OutlierRemovalResults:** Results container

Error Handling

Robust error handling includes: - File I/O error recovery - Memory allocation failure handling - GPU device failure fallback - Numerical stability checks

Logging and Monitoring

Comprehensive logging provides: - Progress tracking for long-running operations - Performance metrics and timing information - Error diagnostics and debugging information - Memory usage monitoring

Applications and Use Cases

Epigenetic Research

MethylCentroid supports various epigenetic research applications:

- **Cancer Epigenetics:** Identification of differentially methylated regions
- **Developmental Biology:** Tracking methylation changes during development
- **Environmental Epigenetics:** Studying methylation responses to environmental factors
- **Population Epigenetics:** Comparing methylation patterns across populations

Quality Control

The outlier detection capabilities enable: - Sample quality assessment - Batch effect identification - Technical artifact detection - Biological outlier identification

Statistical Analysis

Extended centroids support advanced statistical analyses: - Distribution fitting and parameter estimation - Confidence interval calculation - Hypothesis testing frameworks - Bayesian inference methods

Conclusion

MethylCentroid provides a theoretically sound and computationally efficient framework for methylation data analysis. The integration of information-theoretic methods (Jeffreys divergence), probabilistic modeling (Beta distributions), and high-performance computing (GPU acceleration) enables robust and scalable analysis of large-scale methylation datasets.

The mathematical foundations ensure statistical rigor while the computational optimizations enable practical application to real-world genomic datasets. The modular architecture supports both basic centroid calculation and advanced statistical analyses, making MethylCentroid a versatile tool for epigenetic research.

References

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Mathematical Derivations (Appendix)

Derivation of Jeffreys Divergence for Methylation Data

For methylation data, each position represents a Bernoulli trial with success probability x_i . The probability mass function is:

$$P(X_i = 1) = x_i, \quad P(X_i = 0) = 1 - x_i$$

For two methylation profiles $\mathbf{x}$ and $\mathbf{c}$, the Jeffreys divergence becomes:

$$J(\mathbf{x}, \mathbf{c}) = \frac{1}{2} \sum_{i=1}^n [x_i \log(x_i/c_i) + (1-x_i) \log((1-x_i)/(1-c_i)) + c_i \log(c_i/x_i) + (1-c_i) \log((1-c_i)/(1-x_i))]$$

This formulation is equivalent to the symmetrized Kullback-Leibler divergence for Bernoulli distributions.

Beta Distribution Parameter Estimation

The maximum likelihood estimators for Beta distribution parameters can be obtained by solving:

$$\begin{aligned}\psi(\alpha + \beta) - \psi(\alpha) &= (1/N) \sum_{i=1}^N \log x_i \\ \psi(\alpha + \beta) - \psi(\beta) &= (1/N) \sum_{i=1}^N \log(1 - x_i)\end{aligned}$$

where ψ is the digamma function. These equations are solved numerically using optimization algorithms.

This document provides the theoretical foundation for the MethylCentroid software package. For implementation details and usage examples, please refer to the main README.md file and the source code documentation.